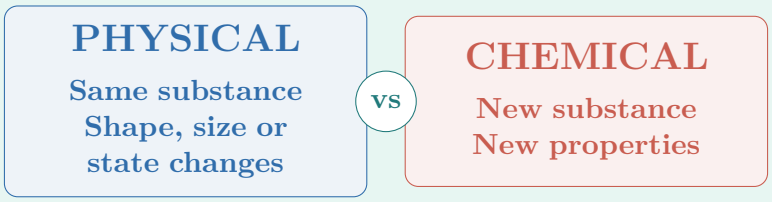
NCERT CURIOSITY • CHAPTER 5

Class 7 Science

# CHANGES AROUND US: PHYSICAL AND CHEMICAL

Textbook solutions ⟶ colourful explanations ⟶ revision




**Your science detective rule**

Always ask: **Has a new substance formed?** Then separately ask: **Can the change be reversed?** These are *different* questions.

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# NCERT textbook questions: complete worked answers

## How to score full marks

For every question about a change, write **(1) the observation, (2) its classification, and (3) whether a new substance forms**. The order below follows *Curiosity*, Chapter 5, textbook pp. 69–71.

## Q1 • Characteristics of a physical change

**Answer. Option (c): statements (i) and (iii).** The state of a material may or may not change, and **no new substance is formed**. Statements (ii) and (iv) describe chemical changes.

## Q2 • Which changes are reversible? (all 10 cases)

| No. | Change | Can it be reversed? |
|---|---|---|
| (i) | Stitching cloth to a shirt | **Usually yes**: remove stitches (small holes may remain). |
| (ii) | Twisting a straight string | **Yes**: untwist it. |
| (iii) | Making idlis from batter | **No**: cooked idli cannot turn back into raw batter. |
| (iv) | Dissolving sugar in water | **Yes, in principle**: evaporate water and recover sugar. |
| (v) | Drawing water from a well | **Yes**: water can be returned, though not necessarily the same portion. |
| (vi) | Ripening of fruits | **No**: unripe fruit cannot be restored. |
| (vii) | Boiling water in an open pan | **Possible in principle**: cool and condense steam; difficult to recover from an open pan because vapour escapes. |
| (viii) | Rolling up a mat | **Yes**: unroll it. |
| (ix) | Grinding wheat grains to flour | **No**: grains cannot be reassembled. |
| (x) | Soil formation from rocks | **No on a practical timescale**: extremely complex long-term changes. |

**Why? Reversible does not always mean chemical or physical.** Grinding is irreversible but mainly physical.

## Q3 • True/False; correct false statements

(i) **True.** Candle wax melts before liquid wax can reach the flame and burn.
(ii) **False.** Condensation of water vapour into liquid water is a **physical** change; the substance remains water.
(iii) **True.** Leaves form compost by decomposition, forming new substances: **chemical change**.
(iv) **True.** Lemon juice and baking soda react and release gas: **chemical change**.

## Q4 • Complete the four statements

(i) Cycle handle: **rusting**; a **chemical** change.
(ii) Folding a handkerchief: **physical**; can be **reversed**.
(iii) Reaction with oxygen and release of heat: **combustion**; a **chemical** change.
(iv) Burnt magnesium forms **magnesium oxide**; it is **basic**; the process is **chemical**.

### Q5 • Water to ice and water to steam

**Answer.** **Both are physical changes.** Freezing and boiling change only the **state** of water: water remains $H_2O$. Ice can melt; water vapour can condense.

### Q6 • Curdling of milk

**Answer.** **Chemical change.** When milk curdles, its components are altered and a curd-like product with different properties forms. It cannot be restored to the original milk by simple physical means.

### Q7 • Formation of soil from rocks

**Answer.** **Both physical and chemical changes may occur.** Wind and flowing water break and carry rock pieces (**physical weathering and erosion**). Water, air and other agents can also alter minerals chemically (**chemical weathering**). Thus soil formation is a long-term combination of processes.

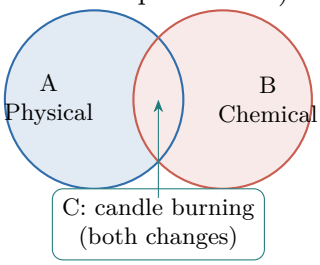
### Q8 • "Eco-friendly Prithvi" story

**Answer.** A suitable new title is **"From Kitchen Waste to a Garden"**. Choose:

- Chopping vegetables, peeling potatoes, cutting fruits: **physical changes**.
- Collecting peels and seeds in a pot: **physical change** (only their location changes).
- Decomposing organic matter to make compost: **chemical change**.
- Germination, growth and flowering: **chemical changes are involved**, along with physical growth; new plant substances form through living processes.

### Q9 • Place each change in Venn diagram A, B or C

**Answer.** **A: physical only** — tearing paper, melting ice, folding clothes.
**B: chemical only** — rusting, curdling milk, ripening fruit, burning magnesium, mixing baking soda with vinegar.
**C: both** — burning a candle (wax melts *and* vapour burns).

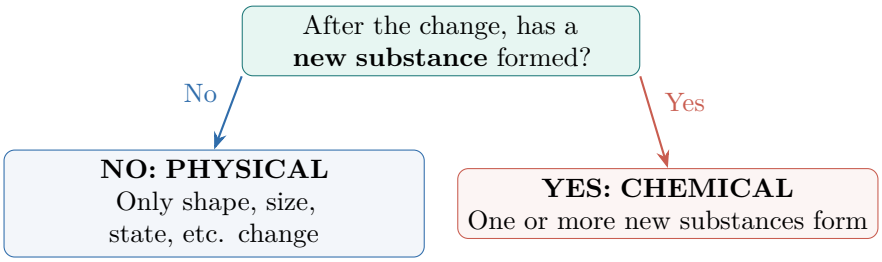



### Q10 • When does lime water turn milky? (Fig. 5.11)

**Answer.** **Setups (a) vinegar + baking soda and (d) lemon juice + baking soda.** In these combinations an acid reacts with baking soda, releasing **carbon dioxide**. The gas turns lime water milky by forming insoluble calcium carbonate.
**Setups (b) lemon juice + vinegar and (c) vinegar + common salt** do not produce this carbon dioxide reaction in the described conditions.

## Understand the chapter with pictures

## 1. A visual decision tree




**Memory trick**

**Physical = form changes. Chemical = substance changes.** But a single event (like a burning candle) may involve *both*.

## 2. Two easy laboratory indicators

**Blow exhaled air through a straw**



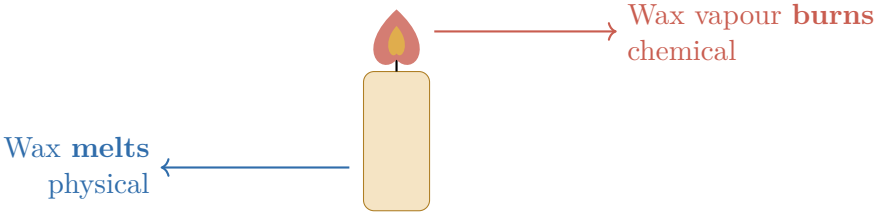
Tap water
Bubbles only

Lime water
Milky / cloudy

**Why milky?**

Exhaled air contains carbon dioxide. In lime water it forms an insoluble white substance (calcium carbonate). **White cloudiness is evidence of a chemical reaction.**

## 3. Candle burning: two changes together




## 4. Weathering versus erosion

| Weathering | Erosion |
|---|---|
| Breaks down or chemically changes rock at its location. | Transports rock fragments/soil through moving wind, water, etc. |
| May be physical or chemical. | Transport itself is a physical process. |

### 5. Combustion needs oxygen and sufficient heat

**Ignition temperature**

The **ignition temperature** is the lowest temperature at which a substance catches fire. Burning involves a chemical reaction with oxygen and releases heat and/or light. A covered candle goes out when its oxygen supply becomes insufficient.

**Desirable and undesirable changes**

**Desirable:** ripening fruits, cooking food, making compost from vegetable peels. **Undesirable:** rusting an iron tool, spoilage of stored food. The *same* change can help in one situation and harm in another: decomposition spoils food, but is helpful for composting.

## Quick comparison and activities

| | Physical change | Chemical change |
|---|---|---|
| Main test | No new substance | New substance(s) form |
| Examples | Ice melts; paper folds; water boils | Iron rusts; food cooks; leaves compost |
| Reversibility | Can be either | Often difficult to reverse |
| Evidence | Changed shape/state | Gas, precipitate, heat/light, odour or colour may be clues (not proof alone) |

**Mini-experiment 1: baking soda plus lemon juice**

**Observe:** fizzing bubbles. **Test:** pass the gas through fresh lime water. **Result:** it turns milky. **Conclude:** a new gas (carbon dioxide) formed, so this is chemical.

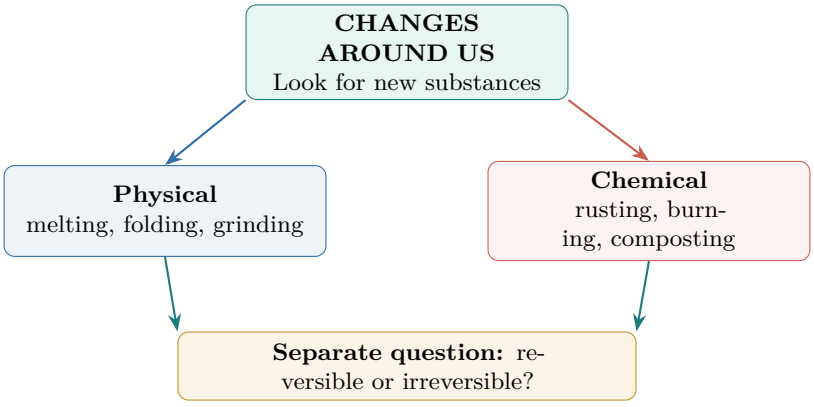
**Mini-experiment 2: candle under a glass**

**Observe:** an uncovered candle keeps burning, while a covered candle soon stops. **Conclude:** oxygen in air supports combustion. A candle stops when the available oxygen becomes insufficient.

**Common mistake**

**Do not say "irreversible means chemical".** Tearing paper and grinding grains are irreversible *physical* changes. Also, bubbles by themselves do not always prove a chemical reaction (blowing air into ordinary water also makes bubbles).

## One-page revision map and exam practice



## Three exam questions to practise

**1.** Why is rusting not a physical change? **2.** Why do we call boiling water a physical change? **3.** Why do a melted candle and a burning candle show different kinds of change?
**Check:** 1. New rust forms. 2. Water remains water. 3. Melting is physical; combustion makes new substances.

## Explore further (textbook project ideas)

A lemon-juice message darkens on warming (adult supervision); observe safe everyday changes while cooking; investigate soil erosion. In the yeast-and-sugar balloon activity, gas inflates the balloon and turns lime water milky because carbon dioxide is released. Chameleon colour changes can be reversible. Do **not** use open flames or a hot iron without adult supervision.

## Classwork challenge: classify and explain

Classify the following as physical, chemical or both, and state whether the change is usually reversible.
1. Crushing chalk into powder
2. Melting a chocolate bar
3. Burning a piece of paper
4. Cooking an idli from batter
5. A wax candle burning steadily
6. A bicycle handle slowly turning reddish-brown

**Suggested answers:** 1. Physical; practically irreversible. 2. Physical; reversible by cooling (though texture may change). 3. Chemical; irreversible. 4. Chemical; irreversible. 5. Both; melting is physical and burning is chemical. 6. Chemical (rusting); hard to reverse.

## Common mistake

A *colour change* alone does not always prove a chemical change. For example, mixing beetroot extract into water changes the colour of the liquid but does not, by itself, establish that a new substance was made. Look for evidence and compare the starting and ending materials.